

H1N1 Vaccine Uptake

An applied EDA case study on demographic and health-related determinants of vaccination behavior

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Project Overview

Objective

This project explores patterns and factors associated with H1N1 flu vaccine uptake using exploratory data analysis (EDA). The goal is to understand demographic and health-related variables that may influence vaccination behavior, working from individual-level survey response data.

26,464

Survey respondents

13

Variables analyzed

R

tidyverse + ggplot2

Data Preparation and Quality Assurance

1

Type Validation

Converted mis-typed numeric fields (age, household composition) and flagged invalid entries as missing rather than silently coercing them.

2

Categorical Standardization

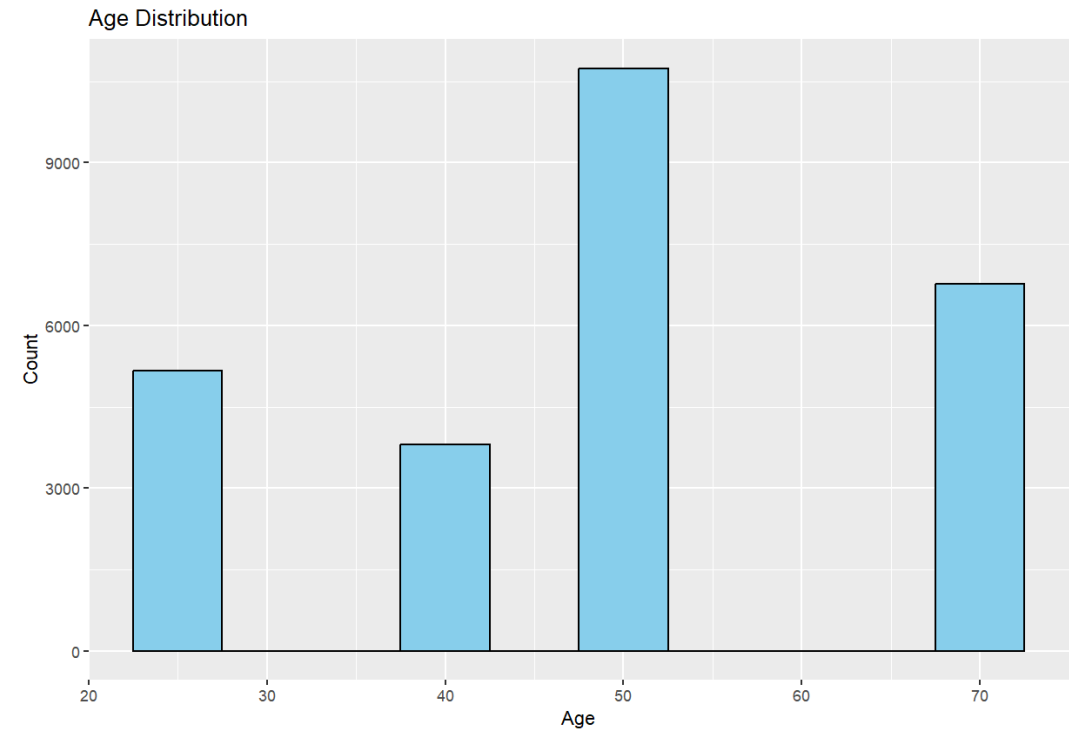
Revealed key indicators (vaccination status, health worker status, chronic condition) as ordered factors for consistent, repeatable analysis.

3

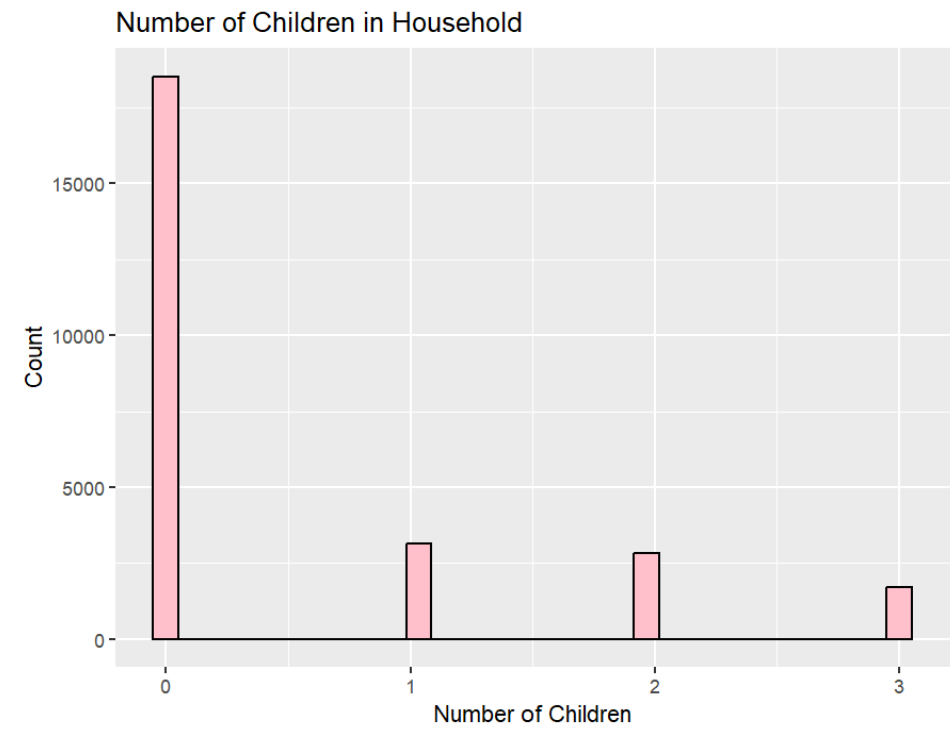
Missing-Data Assessment

Quantified missingness by variable to guide handling: chronic condition (3.5%), health worker status (2.9%), household children (0.9%).

Sample Demographics



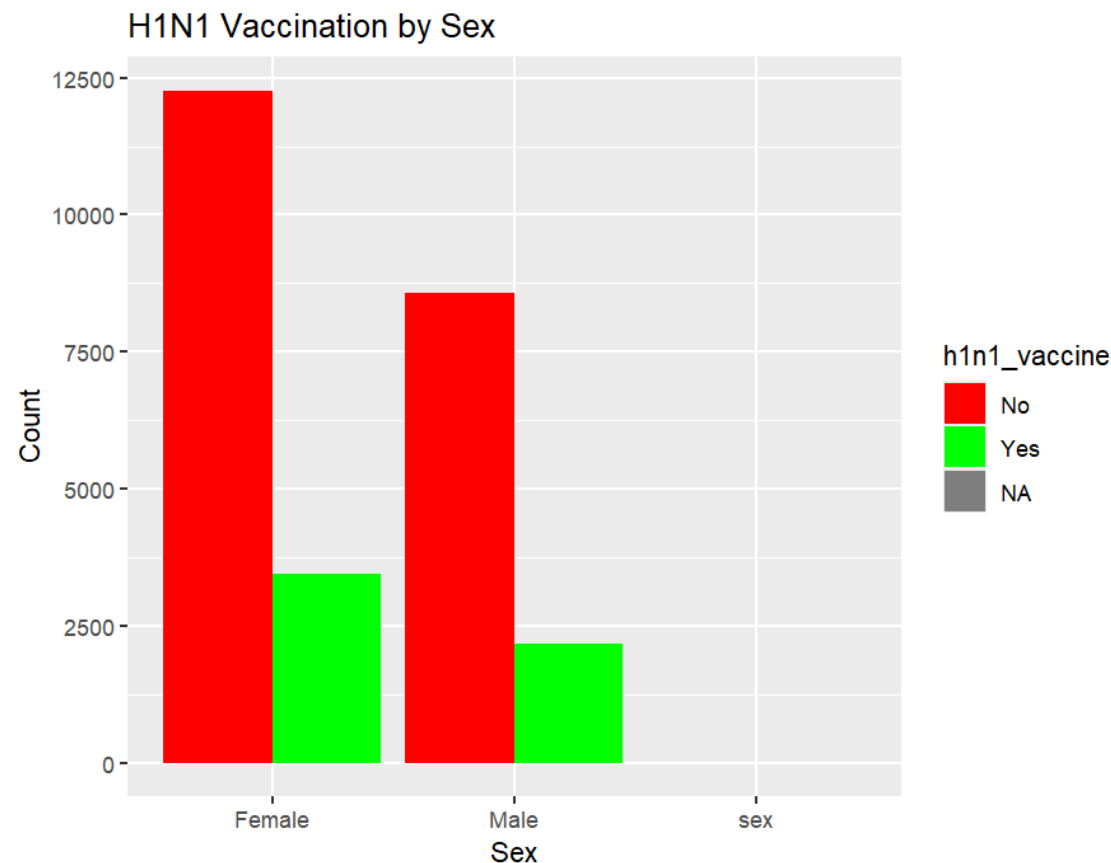
Age distribution (5 recorded groups, incl. missing)



Number of children in household

Age was recorded across five discrete groups (26, 40, 50, and 70 years, plus one missing value). Most households in the sample report no children (median = 0; mean = 0.53).

Vaccination Patterns by Sex



59%

Female respondents

41%

Male respondents

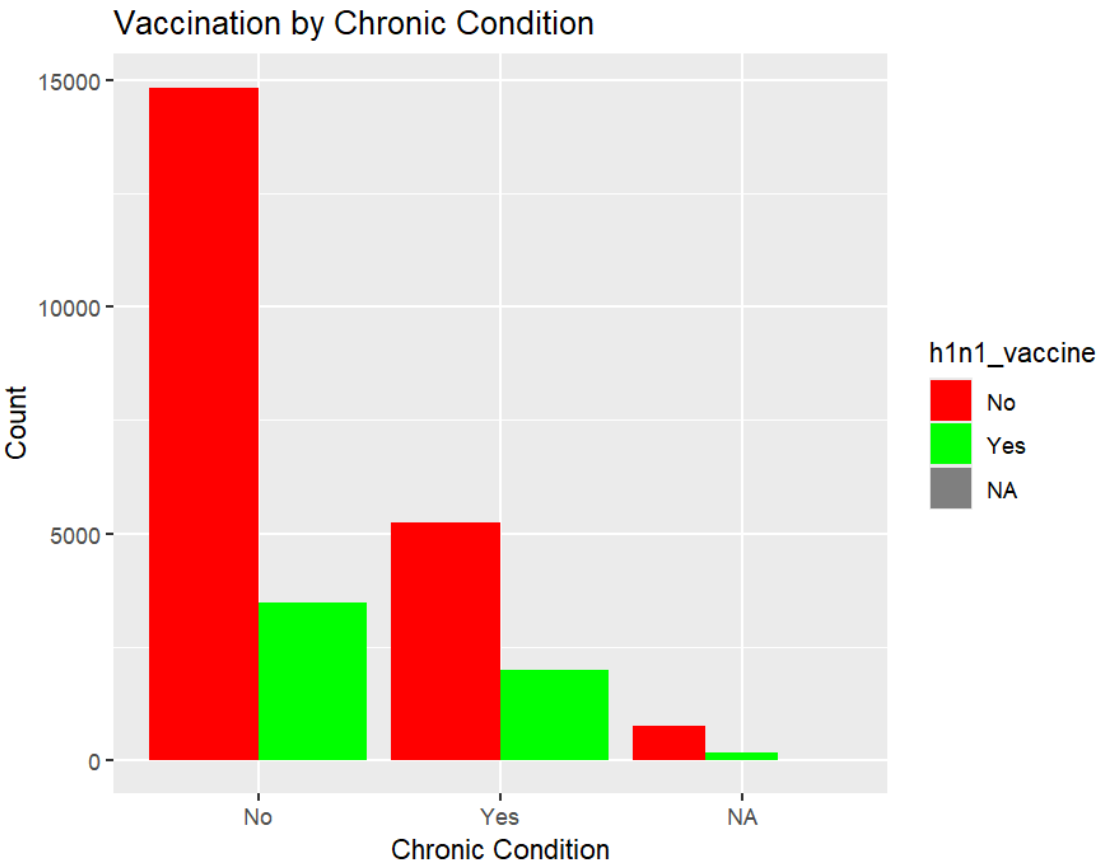
Vaccination uptake differs between male and female respondents, visible in the relative proportions of vaccinated and unvaccinated individuals within each group in the chart at left.

Vaccination Patterns by Chronic Health Status

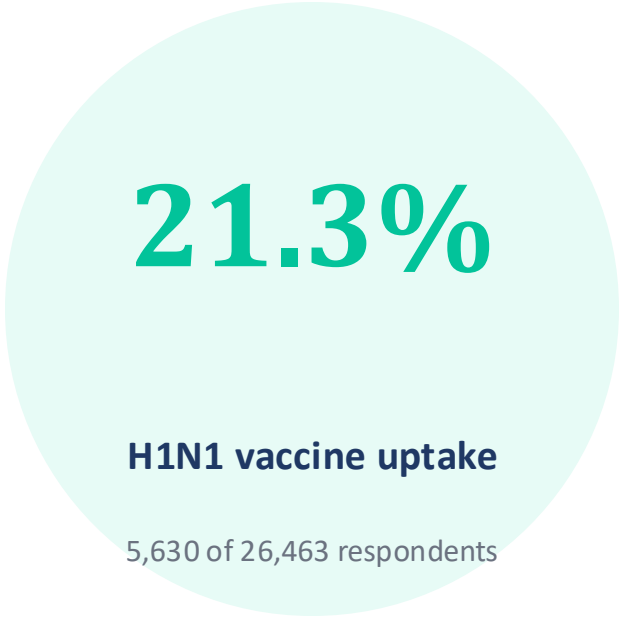
28%

of respondents report a chronic medical condition

Individuals with a chronic medical condition show a higher proportion of vaccination relative to those without, consistent with public health guidance that prioritizes higher-risk groups for vaccination outreach.



Overall Vaccine Uptake

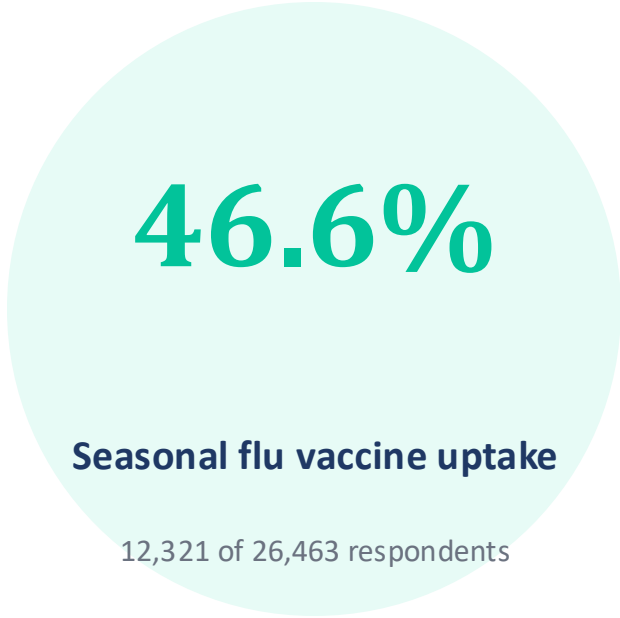


21.3%

This infographic features a light green circle on a white background. Inside the circle, the percentage '21.3%' is written in a large, bold, green font. Below the circle, the text 'H1N1 vaccine uptake' is written in a bold, dark blue font, followed by '5,630 of 26,463 respondents' in a smaller, gray font.

H1N1 vaccine uptake

5,630 of 26,463 respondents



46.6%

This infographic features a light green circle on a white background. Inside the circle, the percentage '46.6%' is written in a large, bold, green font. Below the circle, the text 'Seasonal flu vaccine uptake' is written in a bold, dark blue font, followed by '12,321 of 26,463 respondents' in a smaller, gray font.

Seasonal flu vaccine uptake

12,321 of 26,463 respondents

Key Insights

Key Insights (Preliminary)

- Age distribution shows variation in vaccine uptake across different age groups.
- Sex differences are observed in H1N1 vaccination rates.
- Individuals with chronic medical conditions appear more likely to receive the H1N1 vaccine.
- Household characteristics, such as the number of children, provide additional context for vaccination decisions.